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Developing a mix-specific NDT model for compressive strength prediction of concrete using local Nigerian materials

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ABSTRACT

Accurate estimation of concrete compressive strength using non-destructive testing (NDT) methods remains challenging because widely used empirical correlations often neglect key mix design parameters, particularly the water–cement (w/c) ratio. This study investigates the influence of the w/c ratio on concrete compressive strength and on the predictive performance of combined rebound hammer (RH) and ultrasonic pulse velocity (UPV) models through controlled laboratory experiments using locally sourced Nigerian materials. Six concrete mixes with w/c ratios ranging from 0.45 to 0.70 were prepared with a constant 1:1:2 mix proportion and tested at curing ages of 7, 14, and 28 days. RH and UPV measurements were conducted on laboratory-cured cube specimens and correlated with compressive strength obtained from standard destructive tests. Pearson correlation analysis revealed strong positive relationships between compressive strength and RH ($r = 0.96$) and UPV ($r = 0.93$), while increasing w/c ratio consistently reduced both strength and NDT responses. A multivariate regression model incorporating RH, UPV, and curing age was developed to predict compressive strength across the investigated mixes. The model achieved a coefficient of determination (R^2) of 0.79 with low prediction errors (RMSE = 1.46 MPa; MAE = 1.26 MPa). Comparative analysis showed that the proposed model outperformed single-parameter approaches and highlighted the limitations of generic SonReb correlations when applied without local calibration. The results demonstrate that reliable NDT-based strength prediction is strongly mix-sensitive and requires locally calibrated models for accurate laboratory-based quality control.

1. Introduction

Concrete is the most widely used construction material worldwide owing to its versatility, durability, and cost-effectiveness in infrastructure and building applications [1–6]. A key parameter governing its mechanical performance, particularly the compressive strength, is the water–cement (w/c) ratio. Foundational research has demonstrated that lower w/c ratios generally produce denser cementitious matrices with reduced capillary porosity, leading to higher strength and durability, whereas higher w/c ratios increase the pore connectivity and reduce the mechanical performance [7–10]. Given the importance of compressive strength for structural safety and serviceability, an accurate and timely strength assessment is essential for quality control in both laboratory and field settings [3,7,11–15].

Destructive testing (DT) such as cube or cylinder compression tests is a traditional and widely accepted method for determining the concrete compressive strength [16,17]. Although DT provides standardized results, it is limited to laboratory specimens and extracted cores, making it impractical for widespread use in existing structures or during routine quality control. These limitations have motivated the development of non-destructive testing (NDT) techniques that can estimate strength without damaging the structure. Among the early NDT approaches, RH

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and UPV tests have been widely used owing to their simplicity and cost effectiveness. RH provides an indirect measure of surface hardness, which has been empirically related to compressive strength [18,19], whereas UPV measures the velocity of stress waves through concrete, reflecting its internal density, homogeneity, and the presence of defects [20]. Classic studies have shown that UPV can capture variations in material continuity and microstructure and that the combination of RH and UPV (SonReb methods) can provide more robust estimates of strength than either method alone [21–23].

Recent research has expanded beyond simple SonReb correlations by incorporating UPV into multivariate and data-driven prediction frameworks. For instance, Sathiparan et al. [24] demonstrated that UPV, when combined with material composition parameters and machine learning techniques, can serve as a dominant predictor in cement-stabilized earth systems. Sathiparan et al. [25] applied multivariate models combining UPV and electrical resistivity with mixture characteristics to effectively predict pervious concrete properties, highlighting the sensitivity of UPV to mixed design features. Additional studies integrating UPV with cement content and resistivity for compressive strength prediction further support the view that UPV can improve the accuracy of predictive models when correctly calibrated with the relevant mix parameters [26,27]. These recent studies confirm that UPV has strong potential within predictive frameworks, especially when used in conjunction with other indicators or computational modeling techniques [28–30].

Despite these advances, critical gaps remain in literature. Most existing SonReb correlations and UPV-based models assume mix independence or prioritize statistical fit without systematically evaluating how variations in mix design, particularly the water–cement ratio, influence model performance. Although the influence of the w/c ratio on the compressive strength is well established, its explicit quantitative impact on the accuracy and reliability of multivariate NDT models that combine RH and UPV remains underexplored. This gap is particularly important because NDT calibration curves developed for one set of materials or mix conditions may lead to significant prediction errors when applied to concrete with different w/c ratios or local material sources [31,32].

Moreover, while recent machine learning studies have shown the potential of UPV and auxiliary parameters for property prediction, there are limited studies that specifically quantify the influence of w/c ratio on multivariate NDT model accuracy using locally sourced materials representative of real construction practice. This limitation is particularly relevant in contexts such as Nigeria, where concrete is often produced with locally available aggregates and conventional mix proportions and where the adoption of generic NDT correlations without local calibration can undermine quality control efforts.

To address these gaps, the present study investigates the influence of the water–cement (w/c) ratio on the concrete compressive strength and on the predictive accuracy of multivariate NDT models using locally sourced Nigerian materials. Six concrete mixes with w/c ratios varying from 0.45 to 0.70 were prepared and tested at curing ages of 7, 14, and 28 days. Rebound hammer and ultrasonic pulse velocity measurements were performed on laboratory-cured specimens and correlated with the compressive strength obtained from the standard cube tests. Based on the experimental dataset, a regression-based predictive model integrating RH, UPV, and curing age was developed and evaluated using statistical performance indicators, such as R^2, RMSE, and MAE. The key contribution of this study is the quantification of the effect of the water–cement ratio on the performance of multivariate NDT-based strength prediction models rather than solely on compressive strength outcomes. By calibrating the model using practical mix designs and local Nigerian materials, this study demonstrates how variations in water content systematically influence RH–UPV relationships and prediction accuracy. These findings provide a basis for mix-sensitive NDT calibration and support more reliable concrete quality control in contexts where destructive testing is limited or impractical.

2. Materials and methods

2.1. Experimental workflow

The overall experimental workflow adopted in this study is illustrated in Fig. 1. The procedure comprised material characterization, mix proportioning, specimen casting, curing, non-destructive testing, destructive testing, and statistical analysis. For each curing age (7, 14, and 28 days), UPV testing was conducted first, followed by rebound hammer testing, and finally compressive strength testing. This sequence ensured direct comparability between non-destructive and destructive results for each specimen.

2.2. Materials

The materials used in this study comprised Ordinary Portland Cement (Dangote 3X, 42.5R grade), crushed granite sourced from Abeokuta as coarse aggregate, and natural river sand obtained from Ikorodu as fine aggregate. Potable water was used for both mixing and curing. The physical and chemical properties of the materials are summarized in Tables 1 and 2. The particle size distribution of the fine and coarse aggregates is presented in Fig. 2, confirming compliance with grading requirements for structural concrete.

2.3. Mix proportions

Concrete mixes were designed using a nominal mix ratio of 1:1:2 (cement: fine aggregate: coarse aggregate by weight), consistent with conventional practice for structural-grade concrete. Six mix variants were prepared by varying the w/c ratio from 0.45 to 0.70 while maintaining all other parameters constant. This approach enabled isolation of the influence of water content on both destructive and non-destructive test responses. The detailed mix proportions, water contents, and number of specimens are presented in Table 3.

Notes:

- The material quantities reported are total batch quantities used to cast the indicated number of specimens for each mix.
- The cement: fine aggregate: coarse aggregate ratio (1:1:2 by weight) was kept constant for all mixes.
- Differences in material quantities reflect variations in the number of specimens produced, not changes in mix proportion.

Subsequent mixes were prepared with w/c ratios of 0.50, 0.55, 0.60, 0.65, and 0.70, to cover a wide range of practical and theoretical conditions and assess their influence on workability, strength, and other performance parameters. The precise quantities of cement, fine aggregates, coarse aggregates, and water used in each mixture are listed in Table 3. All the materials were batched by weight and mixed under controlled laboratory conditions to ensure consistency. The variation in the water content was the only adjusted parameter, thereby isolating its influence on the mechanical and physical properties of concrete. This structured approach enabled a comparative analysis of concrete behavior under different w/c ratios, supporting the objective of correlating strength development with both destructive and non-destructive test methods.

2.4. Workability test

The workability of the fresh concrete mixes was evaluated using the slump test in accordance with BS EN 12350-2:2019. The test provided an assessment of the consistency and flowability of each mix as the w/c ratio increased. A schematic illustration and photographic view of the slump test procedure are shown in Fig. 3. As expected, mixes with higher w/c ratios exhibited increased slump values, indicating improved workability with increased water content.

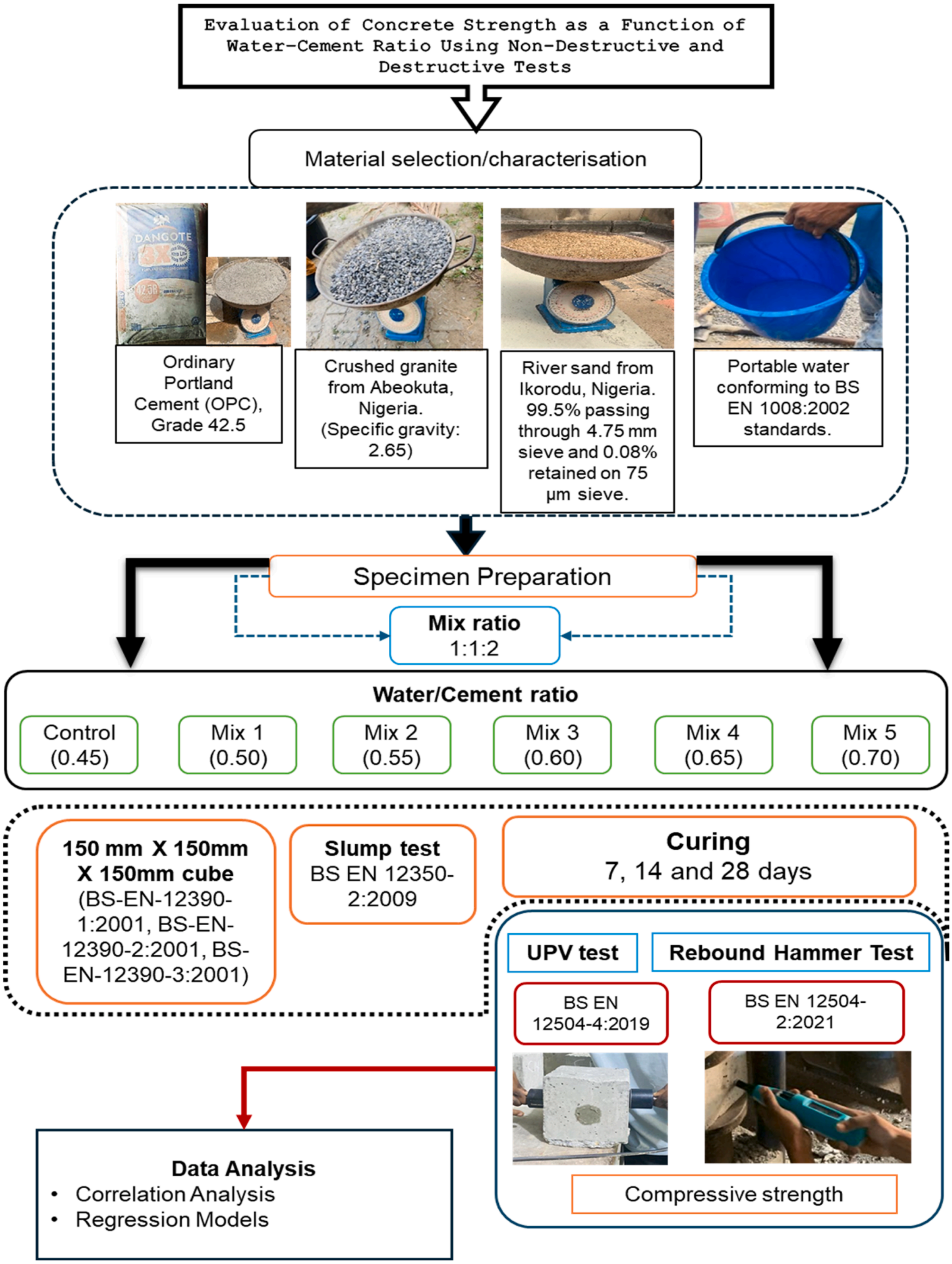

Fig. 1. Schematic flowchart of the experimental workflow adopted in this study.

2.5. *Non-destructive testing*

Non-destructive testing (NDT) was performed using ultrasonic pulse velocity (UPV) and rebound hammer (RH) methods to evaluate internal quality and surface hardness of the concrete specimens. All tests were conducted under controlled laboratory conditions at an ambient temperature of 23 ± 2°C. Prior to testing, instruments were inspected and verified in accordance with the relevant standards to ensure accuracy and repeatability.

2.5.1. *Ultrasonic pulse velocity (UPV) test*

UPV measurements were carried out in accordance with BS EN 12504-4:2019 using the direct transmission method. A schematic illustration of the transducer configuration and the experimental setup are shown in Fig. 4a and b, respectively. Prior to testing, the UPV device was verified using a manufacturer-supplied calibration bar to confirm that measured transit times fell within the specified tolerance range. Coupling gel was applied to ensure adequate acoustic contact between the transducers and the specimen surfaces, thereby minimizing

Table 1
Physical properties of materials used in the experimental program.

Materials	Classification	Property	Value	Standard
Cement (Dangote OPC)	CEM II/A-L 42.5R	Specific gravity	3.15	EN 197-1
Fine aggregate	Natural river sand	Specific gravity	2.57	ASTM C128 [33]
		Fineness modulus	2.6	ASTM C136 [34]
Coarse aggregate	Crushed granite	Specific gravity	2.65	ASTM C127 [35]

Table 2
Physicochemical properties of fine aggregate.

S/N	Parameter	Results
1	pH 28°C (10% solution)	7.2
2	Conductivity (Ms/cm)	0.47
3	Total Dissolved Solid (DS)	340
4	Chloride ion (Cl^-) @ 250 mg/l (max)	0.292
5	Sulphate (SO_3^{2-}) @ 250 mg/l (max)	0.12
6	Organic matter (TOC X 1.72)	0.774

interface-related signal attenuation. For each cube specimen, the transmitting and receiving transducers were positioned on opposite faces to allow direct wave propagation through the full specimen thickness. The path length was measured using a digital caliper (accuracy ±0.01 mm), and the transit time was recorded automatically by the equipment. To ensure repeatability, a minimum of three measurements were taken per specimen at slightly adjusted positions along the same transmission axis. The reported UPV value corresponds to the arithmetic mean of these readings, and the associated standard deviation was calculated to quantify measurement variability [37]. The ultrasonic pulse velocity was calculated using Eq. 1.

$$V = \frac{L}{T} \tag{1}$$

Where $V=$ ultrasonic pulse velocity (m/s), $L=$ path length of the pulse through the specimen (m), $T=$ measured transit time (s). Measurement uncertainty in UPV testing may arise from variations in coupling conditions, minor differences in transducer alignment, and intrinsic concrete heterogeneity.

2.5.2. Rebound hammer (RH) test

Rebound hammer testing was conducted in accordance with BS EN 12504-2:2021 using a Proceq Original Schmidt hammer. The schematic representation and experimental setup are presented in Fig. 5a and b respectively. Before testing, the rebound hammer was checked using a standard steel calibration anvil to verify that rebound readings fell within the manufacturer's specified calibration limits. This procedure ensured compliance with operational accuracy requirements.

Each specimen was placed on a compression testing machine to provide rigid support and prevent movement during impact. The hammer was applied perpendicular to the test surface to maintain consistent impact conditions. Ten rebound impacts were performed at different locations on each specimen surface, with sufficient spacing between test points to prevent interaction effects. The mean rebound quotient (Q) was calculated from the valid readings, and the standard deviation was determined to assess variability and repeatability. The empirical relationship between rebound quotient and compressive strength follows the general form given in Eq. 2.

$$f_c = aQ + b \tag{2}$$

Where $f_c=$ estimated compressive strength of concrete (MPa), $Q=$ mean rebound quotient, $a$and $b=$ empirical constants dependent on material properties and calibration. Potential sources of uncertainty in RH measurements include surface moisture condition, local aggregate distribution beneath the impact point, carbonation effects, and operator alignment.

Table 3
Concrete mix proportions and specimen details for different water–cement (w/c) ratios.

Materials	Concrete Specimen Identification					
	Control	Mix-1	Mix-2	Mix-3	Mix-4	Mix-5
Cement (Kg)	233.89	140.33	140.33	140.33	140.33	140.33
Fine aggregate (Kg)	233.89	140.33	140.33	140.33	140.33	140.33
Coarse aggregate (Kg)	467.78	280.67	280.67	280.67	280.67	280.67
Water–cement ratio (w/c)	0.45	0.50	0.55	0.60	0.65	0.70
Water (Kg)	105.25	70.2	77.2	84.2	91.2	98.2
Total No. of Cubes	15	9	9	9	9	9

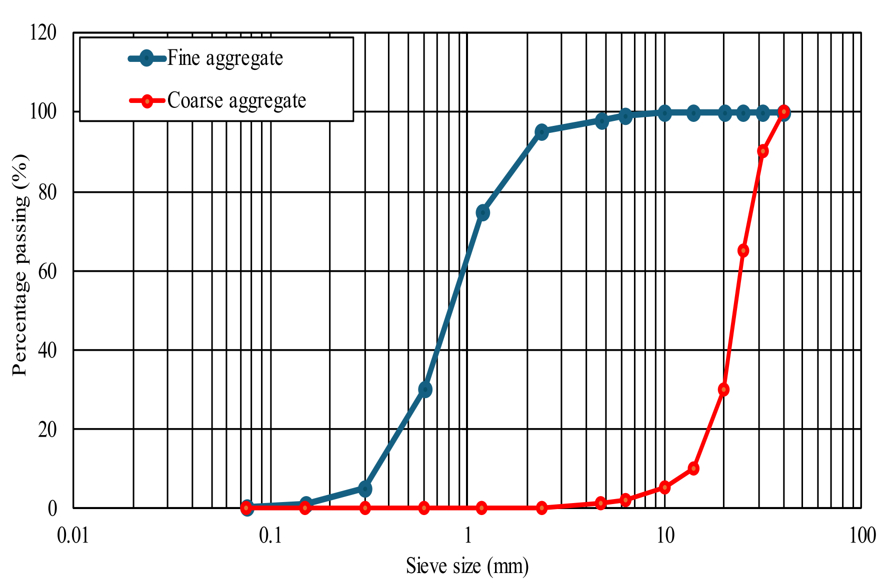

Fig. 2. Particle size distribution of fine and coarse aggregates used in the concrete mixes.

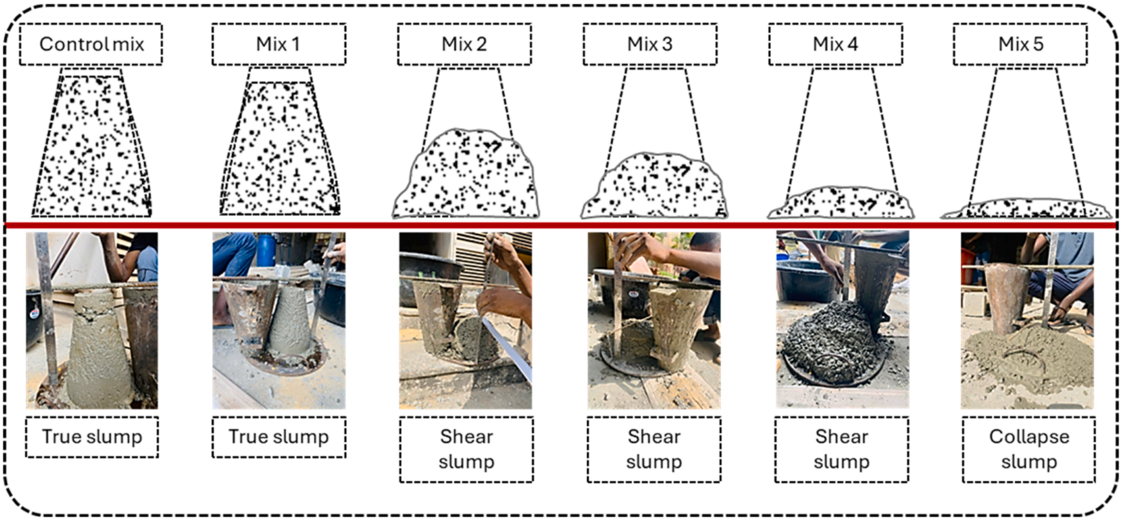

Fig. 3. Schematic illustration and photographic view of the slump test procedure.

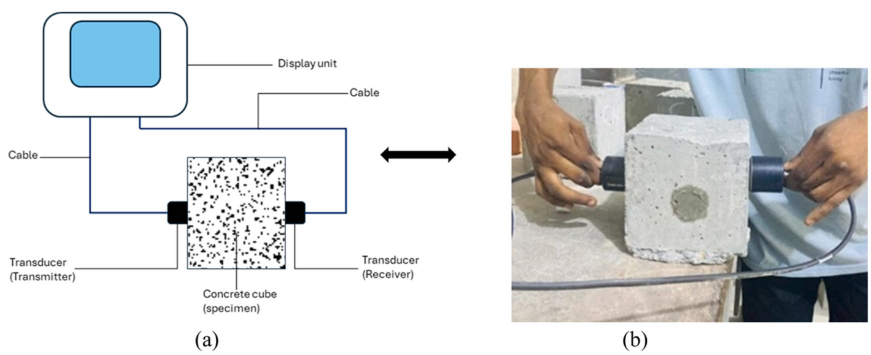

Fig. 4. UPV testing setup: (a) schematic illustration of the direct transmission method; (b) photograph of the test configuration during measurement.

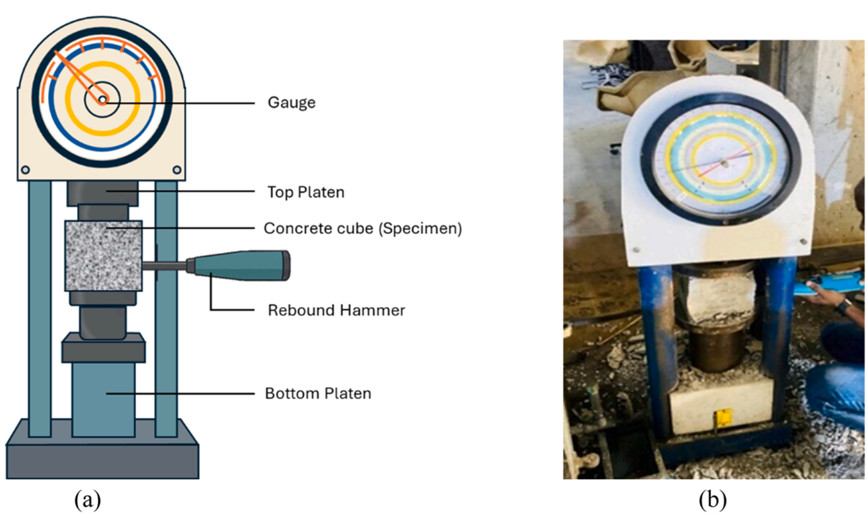

Fig. 5. Rebound hammer testing of concrete cube specimens: (a) schematic illustration of the test principle; (b) photograph of the test setup during measurement.

2.6. Destructive compressive strength test

The destructive compressive strength of the concrete specimens was determined using a 1000 kN capacity compression testing machine. Each cube specimen was subjected to uniform axial loading until failure, and the compressive strength was calculated as the ratio of the maximum applied load to the cross-sectional area [36]. The destructive test results served as reference values for validating the non-destructive testing measurements and for developing predictive regression models. The specimens were subjected to uniform loading until failure, and the compressive strength was calculated using Eq. 3.

$$f_{cu} = \frac{P}{A} \quad (3)$$

where f_{cu} = compressive strength (MPa), P is the maximum applied load (N), and A is the cross-sectional area of a cube (mm^2). The results were averaged for each group and correlated with NDT values for further analysis

3. Results and discussion

3.1. Effect of water–cement ratio on mechanical and NDT responses

The experimental results demonstrate a clear and systematic influence of the w/c ratio on workability, mechanical performance, and non-destructive testing (NDT) responses of concrete across all curing ages. As summarized in Table 4 and illustrated in Figs. 6–8, increasing the w/c ratio generally resulted in higher slump values, indicating improved workability due to increased free water content. The Slump Difference in Table 4 represents the incremental change in slump value (mm) relative to the immediately preceding w/c ratio mix. Although slump values exhibited an overall increasing trend with increasing w/c ratio, a slight reduction was observed at w/c = 0.65 compared to w/c = 0.60, resulting in a negative Slump Difference. This deviation is attributed to experimental variability and possible segregation or instability effects at higher water contents, which can produce non-uniform deformation behavior during the slump test despite the general increase in workability. Compressive strength consistently decreased with increasing w/c ratio at 7, 14, and 28 days. Mixes with lower w/c ratios (0.45 and 0.50) exhibited higher early- and later-age strength development, whereas higher w/c ratios ($\geq$ 0.65) resulted in reduced strength due to increased capillary porosity and a less dense cementitious matrix, consistent with Abrams' law and porosity-based strength theory.

RH measurements followed a similar trend. The *Q* increased with curing age for all mixes, reflecting progressive hydration and surface hardening. However, for a given curing age, *Q* values decreased systematically with increasing w/c ratio, indicating reduced surface hardness associated with higher pore volume. Minor non-linear variations observed at intermediate ages for higher w/c ratios may be attributed to delayed hydration and moisture redistribution effects in more porous mixes. UPV results also showed strong dependence on w/c ratio. Higher UPV values were recorded for mixes with lower w/c ratios, reflecting greater internal density and material continuity. Conversely, increasing the w/c ratio resulted in lower UPV values across all curing ages, indicating reduced wave transmission efficiency due to increased pore connectivity. Although UPV generally increased with curing age, the rate of increase diminished at higher w/c ratios, suggesting limited microstructural refinement in mixes containing excess water. Across all mixes and curing ages, compressive strength exhibited standard deviations between 0.65–1.85 MPa (COV: 2.5–6.8%), rebound quotient between 0.9–2.1 units (COV: 2–5%), and UPV between 0.03–0.09 km/s (COV: 0.8–2.5%). These low COV values indicate acceptable experimental repeatability and confirm that observed strength variations are primarily attributable to mix design effects rather than measurement inconsistency [36–39].

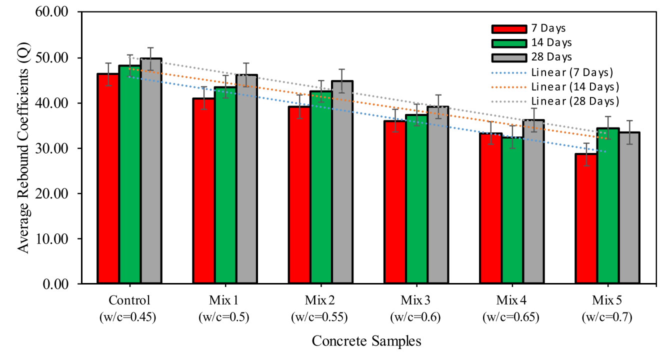

Fig. 6. Effect of water–cement ratio on rebound quotient (Q) at different curing ages.

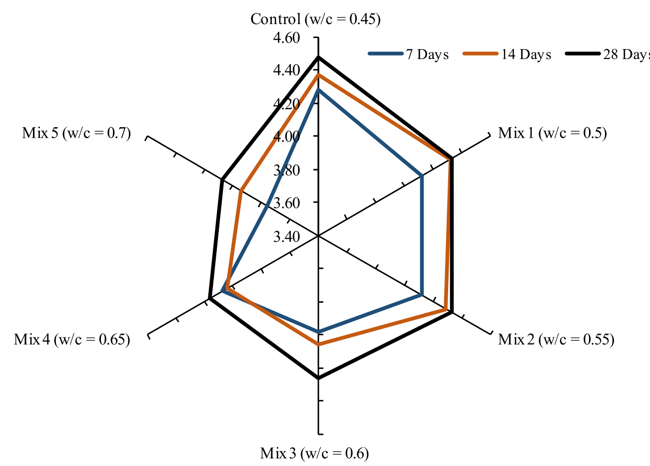

Fig. 7. Effect of water–cement ratio on ultrasonic pulse velocity (UPV) at different curing ages.

Table 4
Effect of *w/c* ratio on slump, *fcu*, rebound quotient (Q), and UPV at different curing ages (Mean ± SD).

w/c	Slump (mm)	Slump Difference (mm)	***fcu*** (MPa)			rebound quotient (Q)			UPV (km/s)		
			7days	14days	28days	7days	14days	28days	7days	14days	28days
0.45	11 ± 1.5	—	27.64 ± 0.78	30.98 ± 0.92	31.07± 0.85	46.34 ± 1.12	48.18 ± 1.25	49.68 ±1.18	4.28± 0.05	4.37± 0.04	4.48± 0.06
0.50	30 ± 2.1	+19	21.04 ± 0.82	29.19 ± 1.03	31.63 ± 0.94	41.00 ± 1.36	43.53 ± 1.22	46.20 ± 1.41	4.12 ± 0.06	4.32 ± 0.05	4.33 ± 0.05
0.55	148 ± 3.5	+118	22.81 ± 0.91	24.67 ± 0.88	25.56 ± 0.96	39.10 ± 1.28	42.53 ± 1.30	44.83 ± 1.26	4.12 ± 0.05	4.29 ± 0.06	4.33 ± 0.05
0.60	189 ± 4.0	+41	19.70 ± 0.73	20.96 ± 0.81	24.44 ± 0.89	36.03 ± 1.14	37.30 ± 1.18	39.07 ± 1.21	3.98 ± 0.07	4.06 ± 0.05	4.26 ± 0.06
0.65	168 ± 3.2	−21	19.04 ± 0.85	15.11 ± 0.79	19.93 ± 0.92	33.27 ± 1.09	32.37 ± 1.15	36.13 ± 20	4.07 ± 0.06	4.04 ± 0.06	4.16 ± 0.05
0.70	210 ± 4.6	+42	12.22 ± 0.68	16.44 ± 0.83	17.33 ± 0.74	28.60 ± 1.04	34.40 ± 1.19	33.40 ± 1.17	3.76 ± 0.08	3.94 ± 0.07	4.07 ± 0.06

*Note: *Slump Difference* represents the incremental change in slump value (mm) relative to the immediately preceding w/c ratio mix. Negative values indicate a reduction in measured slump compared to the previous mix, attributed to experimental variability and potential segregation or instability effects at higher water contents.

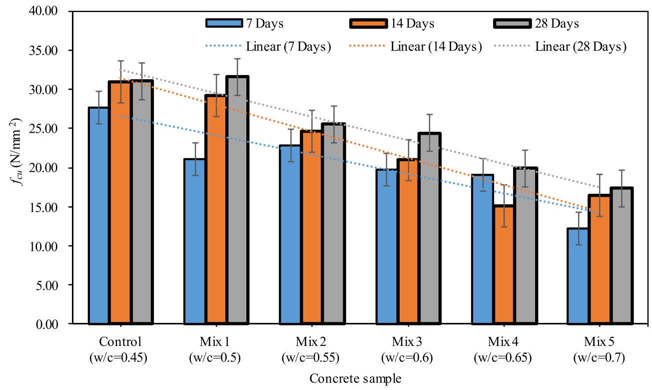

Fig. 8. Effect of w/c ratio on compressive strength at different curing ages.

3.2. Correlation analysis

Pearson correlation analysis was conducted to quantify the relationships among f_{cu}, Q, UPV, water–cement ratio, and slump. The Pearson correlation coefficient, r, between two variables x and y was calculated using Eq. 4.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{4}$$

Where $x_i$and $y_i$are paired observations, $\bar{x}$and $\bar{y}$are their respective means, and $n$is the number of observations. The resulting correlation matrix (Fig. 9) reveals statistically significant trends consistent with established concrete behavior. The w/c ratio exhibited a strong negative correlation with compressive strength (r = −0.89), confirming the detrimental effect of increased water content on mechanical performance. Similarly, strong negative correlations were observed between the w/c ratio and Q (r = −0.94) and between the w/c ratio and UPV (r = −0.81). Both NDT parameters showed strong positive correlations with compressive strength, with r = 0.96 for Q and r = 0.93 for UPV. The high inter-correlation between Q and UPV (r = 0.92) indicates complementary sensitivity to concrete quality while also highlighting the potential for multicollinearity, which is addressed through error-based model evaluation.

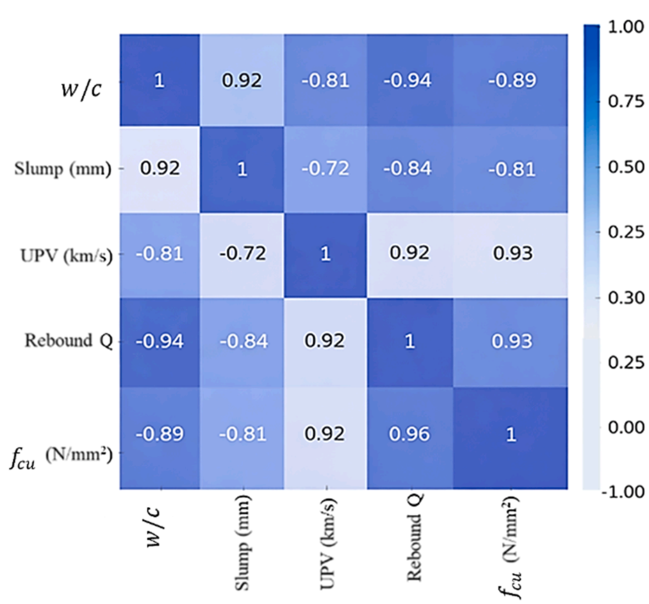

Fig. 9. Pearson correlation heatmap of key concrete performance parameters.

3.3. Multivariate regression modelling for compressive strength prediction

Based on the strong correlations identified in Section 4.2, regression analysis was employed to develop predictive relationships between compressive strength and NDT parameters. The general multivariate regression framework adopted in this study is expressed in Eq. 5.

$$f_{cu} = \alpha_0 + \alpha_1 Q + \alpha_2 V + \alpha_3 t \tag{5}$$

Where f_{cu} is the compressive strength, Q is the rebound quotient, $V$is the ultrasonic pulse velocity, $t$is the curing age, and α_0–α_3 are regression coefficients. Eq. 6 represents a conceptual regression form and does not correspond to a calibrated predictive model; therefore, it is not associated with a specific coefficient of determination or error indices. Using the experimental dataset encompassing all w/c ratios and curing ages, regression coefficients were estimated to derive a calibrated strength prediction model. This resulted in the final regression equation proposed in this study.

$$f_{cu} = 23.31 + \left[3.686\frac{(Q40.21)}{6.273}\right] + \left[2.073\frac{(V4.182)}{0.1831}\right] \tag{6}$$

The final calibrated multivariate model, integrating rebound hammer and UPV measurements to capture surface hardness and internal material continuity effects. The predictive performance of Eq. 7-Eq. 9 was evaluated using the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE), defined respectively as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{7}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{8}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \tag{9}$$

Where $y_i$and $\hat{y}_i$are the measured and predicted compressive strengths, respectively. Eq. 7 achieved R^2 = 0.79, with low prediction errors (RMSE = 1.46 MPa; MAE = 1.26 MPa), indicating strong practical accuracy across the investigated w/c ratios and curing ages. These error values indicate strong practical accuracy across the dataset, especially compared to single-parameter correlations. The predicted–measured comparisons at 7, 14, and 28 days (Fig. 10a–c) show that the model provides close agreement with destructive test results across ages, with deviations increasing slightly at higher w/c ratios. This observation supports the conclusion that mixes with higher water contents introduce greater variability in both strength and NDT readings due to higher porosity and weaker matrix continuity, which can reduce model stability. The 3D response surface (Fig. 11) further illustrates that compressive strength increases with both Q and UPV, confirming that combining the two NDT indicators enhances sensitivity to concrete quality compared to using either metric alone.

3.4. Comparative evaluation of strength prediction methods

Table 5 compares the predictive performance of different compressive strength estimation methods evaluated in this study. The rebound hammer (RH)–only approach showed a relatively high coefficient of determination (R^2 = 0.90); however, it produced comparatively larger prediction errors (RMSE = 3.94 MPa; MAE = 3.49 MPa), indicating systematic underestimation of strength and sensitivity to surface conditions. This behavior is consistent with the surface-dependent nature of rebound hammer measurements and their susceptibility to moisture and finishing effects. The UPV-based empirical model also exhibited strong correlation with compressive strength (R^2 = 0.87) but generated higher

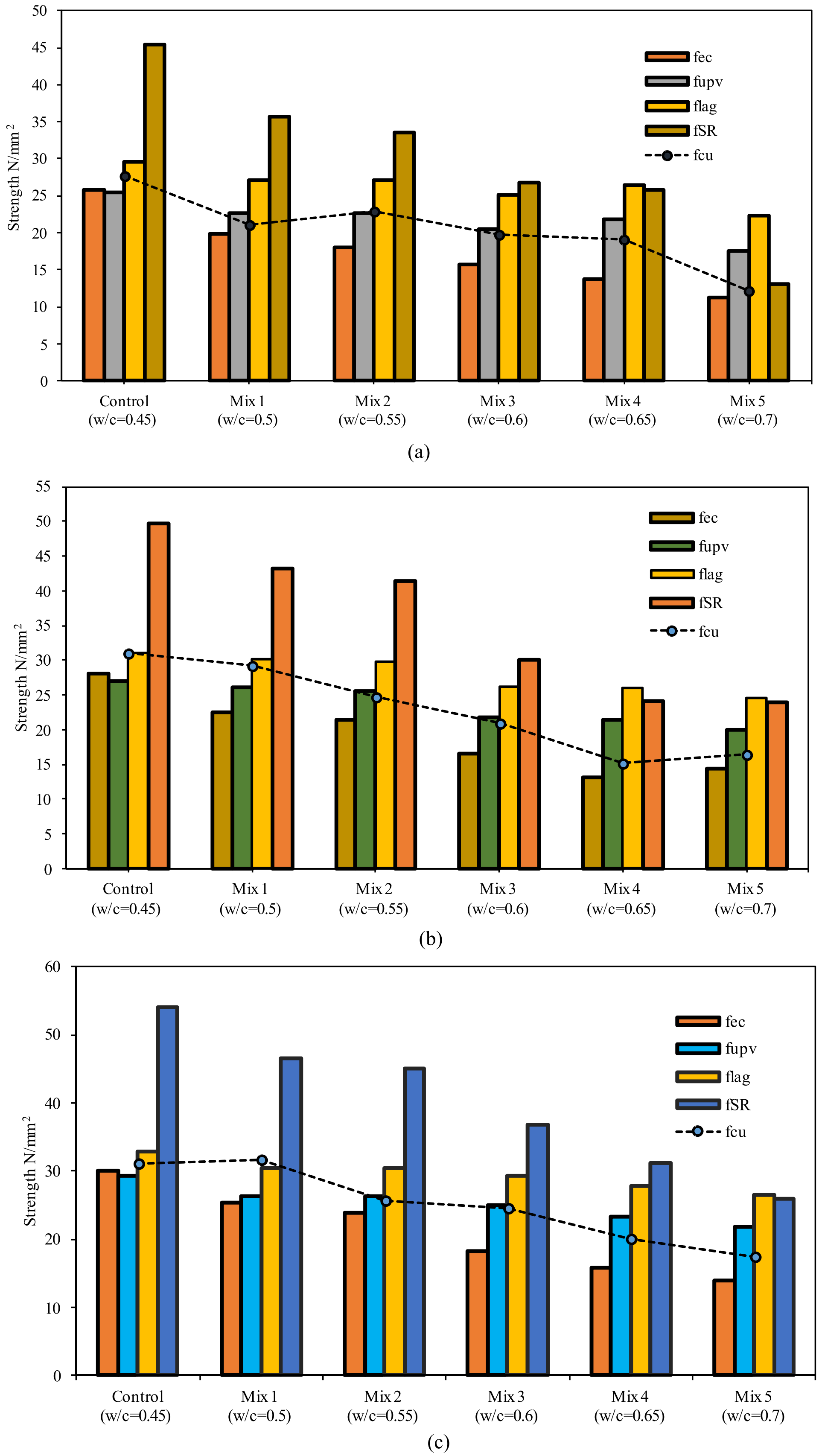

Fig. 10. Comparison of predicted and measured compressive strength:(a) 7 days;(b) 14 days; and (c) 28 days.

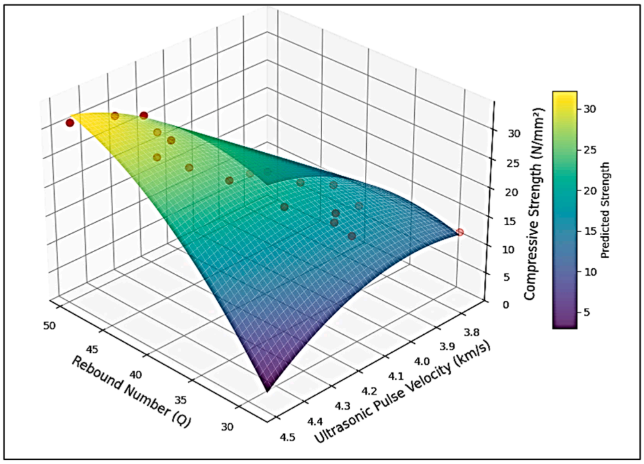

Fig. 11. Three-dimensional response surface illustrating the combined effect of *Q* and *UPV* on concrete *fcu*.

Table 5
Predictive performance of different strength estimation methods.

Prediction method	R^2	RMSE (MPa)	MAE (MPa)	Notes
Rebound hammer (Q-curve)	0.90	3.94	3.49	Consistently underestimates strength
UPV (empirical model)	0.87	3.21	2.64	Overestimates at high w/c ratios
SonReb (RILEM)	0.93	13.48	12.36	Large magnitude errors for local materials
Proposed model (Eq. 6)	0.79	1.46	1.26	Best overall accuracy across dataset (mix-sensitive)

prediction errors (RMSE = 3.21 MPa; MAE = 2.64 MPa) compared to the proposed multivariate model. Notably, the UPV-only model systematically overestimated compressive strength at higher water–cement ratios (w/c $\geq$ 0.65), despite UPV values decreasing with increasing water content. This apparent inconsistency does not arise from the UPV measurements themselves, which accurately reflect reductions in internal density and material continuity, but rather from limitations of the generic, uncalibrated empirical UPV–strength relationship employed. Such models do not account for mix-dependent effects, including increased capillary porosity, delayed hydration, and altered stiffness development at higher water contents, leading to inflated strength estimates when applied outside their original calibration range.

Although the generic SonReb (RILEM) formulation produced a high coefficient of determination (R^2 = 0.93), it resulted in very large prediction errors (RMSE = 13.48 MPa; MAE = 12.36 MPa) for the locally sourced Nigerian materials investigated. This finding demonstrates that high R^2 values may indicate trend agreement without ensuring accurate magnitude prediction, particularly when empirical models are transferred across different material systems without recalibration. In contrast, the proposed mix-specific multivariate regression model, which integrates rebound hammer response, ultrasonic pulse velocity, and curing age, achieved the lowest prediction errors (RMSE = 1.46 MPa; MAE = 1.26 MPa) with a coefficient of determination of R^2 = 0.79. The improved accuracy of this model highlights the advantage of combining complementary NDT indicators and explicitly accounting for mix sensitivity through local calibration. Overall, this comparison confirms that reliable concrete strength estimation using NDT methods requires calibration to local materials and mix designs, and that reliance on generic, uncalibrated empirical correlations can lead to significant prediction errors, particularly at higher water–cement ratios [40,41].

3.5. Correction factor for rebound-based strength estimation

To improve the reliability of rebound hammer–based compressive strength estimation, correction factors were derived by comparing strengths predicted using rebound hammer calibration curves with those obtained from standard destructive compression tests conducted under controlled laboratory conditions. The specimen-level and average correction factors were computed using Eq. 10 and Eq. 11, respectively, and the resulting values are summarized in Table 6.

$$CF_i = \frac{f_{cu,DT,i}}{f_{cu,RH,i}} \tag{10}$$

$$CF = \frac{1}{n}\sum_{i=1}^{n} CF_i \tag{11}$$

where $f_{cu,DT}$ is the compressive strength obtained from destructive testing and $f_{cu,RH}$ is the strength estimated from the rebound hammer calibration curve and n is total numbers. The calculated correction factors exhibit noticeable variability across different water–cement ratios and curing ages, reflecting the sensitivity of rebound hammer measurements to concrete mix characteristics and microstructural development. In general, concretes produced with higher water–cement ratios required larger correction factors, indicating a greater tendency for rebound hammer measurements to underestimate compressive strength in more porous and heterogeneous matrices. Conversely, lower water–cement ratio mixes showed correction factors closer to unity, consistent with denser microstructures and more reliable rebound-based estimates. Based on the experimental dataset, an average correction factor of 1.19 was obtained. Application of this factor reduced systematic underestimation associated with rebound hammer–based strength predictions for the concretes investigated. However, this correction factor is applicable only to concrete produced with materials, mix proportions, and curing conditions similar to those examined in this study. Direct application to in-situ concrete or to mixes incorporating different materials or exposure conditions should therefore be preceded by appropriate local calibration and validation [42,43]. The results confirm that rebound hammer–based strength estimation is inherently mix-sensitive and that reliance on generic calibration curves can introduce significant prediction bias. Locally derived correction factors, established under controlled laboratory conditions, provide a practical means of improving agreement between rebound-based estimates and reference compressive strength values when destructive testing is limited.

Conclusion

This study examined the influence of the w/c ratio on concrete compressive strength and on the predictive performance of NDT–based strength estimation using controlled laboratory experiments with locally sourced Nigerian materials. Six concrete mixes with w/c ratios between 0.45 and 0.70 were evaluated at curing ages of 7, 14, and 28 days using RH, UPV and standard cube compression tests. The results confirmed that increasing the w/c ratio systematically reduced compressive strength, rebound quotient, and ultrasonic pulse velocity, consistent with increased porosity and reduced matrix continuity. Strong correlations were observed between compressive strength and RH (r = 0.96) and between compressive strength and UPV (r = 0.93), demonstrating the sensitivity of both surface-based and internal NDT indicators to mix design parameters. However, prediction deviations increased at higher w/c ratios, indicating that concrete heterogeneity associated with excess water content adversely affects NDT-based strength estimation.

A locally calibrated multivariate regression model integrating RH, UPV, and curing age achieved low prediction errors (RMSE = 1.46 MPa; MAE = 1.26 MPa) and outperformed single-parameter NDT approaches and a generic SonReb correlation when applied to the investigated

Table 6
Correction factor for rebound hammer–based compressive strength estimation.

Sample	(Q)			Compressive Strength from Q curve (N/mm^2)			Compressive Strength from DT (N/mm^2)			Compressive Strength Difference (N/mm^2)			Compressive Strength Average Difference (N/mm^2)	Factor			Average factor
	Curing ages (Days)			Curing ages (Days)			Curing ages (Days)			Curing ages (Days)				Curing ages (Days)			
	7	14	28	7	14	28	7	14	28	7	14	28		7	14	28	
Control	46.34	48.18	49.68	25.8	28	30	27.64	30.98	31.07	1.84	2.98	1.07	1.96	1.07	1.11	1.04	1.07
Mix-1	41	43.53	46.2	19.83	22.5	25.33	21.04	29.19	31.63	1.21	6.69	6.3	4.73	1.06	1.30	1.25	1.20
Mix-2	39.1	42.53	44.83	18	21.33	23.83	22.81	24.67	25.56	4.81	3.34	1.73	3.29	1.27	1.16	1.07	1.17
Mix-3	36.03	37.3	39.07	15.67	16.5	18.17	19.7	20.96	24.44	4.03	4.46	6.27	4.92	1.26	1.27	1.35	1.29
Mix-4	33.27	32.37	36.13	13.67	13.17	15.83	19.04	15.11	19.93	5.37	1.94	4.1	3.80	1.39	1.15	1.26	1.27
Mix-5	28.6	34.4	33.4	11.17	14.33	13.83	12.22	16.44	17.33	1.05	2.11	3.5	2.22	1.09	1.15	1.25	1.16
														Average factor for any day			1.19

material system. Although the generic SonReb formulation showed a high coefficient of determination, it produced large magnitude errors, highlighting that statistical correlation alone does not ensure accurate strength prediction without local calibration.

To further improve rebound-based estimates within the experimental dataset, an average correction factor of 1.19 was derived. Application of this factor reduced systematic underestimation relative to laboratory destructive test results for concretes produced with similar materials and mix proportions. The proposed model and correction factor are therefore applicable to laboratory-based quality control of newly produced concrete where mix proportions and curing conditions are known. Overall, the findings demonstrate that reliable NDT-based prediction of concrete compressive strength requires mix-sensitive and locally calibrated models rather than reliance on generic empirical correlations. While the study provides a robust laboratory framework for improving strength estimation, further validation using field data is required before extending the proposed approach to in-situ concrete assessment.

Limitations and future work

The findings of this study should be interpreted considering several limitations. First, all experiments were conducted under controlled laboratory conditions using standard-cured concrete specimens. Consequently, the results do not fully capture field-related influences such as moisture gradients, surface carbonation, temperature variation, reinforcement presence, and in-situ stress conditions, all of which can affect rebound hammer and ultrasonic pulse velocity responses.

Second, the experimental program focused on a limited range of mix designs based on a conventional 1:1:2 proportion, with the water–cement ratio as the primary variable. While this approach enabled isolation of the w/c ratio effect, it does not account for the influence of supplementary cementitious materials, chemical admixtures, or alternative aggregate types commonly used in practice. Third, the regression model was developed using a relatively modest dataset derived from laboratory-cured specimens, and its predictive performance was evaluated within the same material system. As a result, the generalizability of the model to other material sources or construction environments may be limited without further validation.

Future research should extend the proposed framework through broader experimental and field-based validation. Incorporating additional mix design variables, such as supplementary cementitious materials, admixtures, and recycled or lightweight aggregates, would improve the robustness of NDT-based prediction models. Long-term monitoring at extended curing ages (e.g., 56 and 90 days) is recommended to evaluate the stability of NDT–strength relationships beyond early-age hydration. Field investigations on existing structures, accounting for carbonation depth, moisture condition, and reinforcement effects, would further enhance the applicability of the model. In addition, integrating advanced statistical or machine-learning approaches with larger datasets may improve predictive accuracy while preserving interpretability. These future developments would support more reliable, mix-sensitive, and region-specific non-destructive strength assessment frameworks for concrete structures.

Abbreviations

ACI	American Concrete Institute
C–S–H	Calcium Silicate Hydrate
CH	Calcium Hydroxide
CTM	Compression Testing Machine
DT	Destructive Testing
EN	European Norm (standard)
f_{cu}	Compressive Strength (cube)
NDT	Non-Destructive Testing
OPC	Ordinary Portland Cement
Q	Rebound Quotient
RH	Rebound Hammer
SCM	Supplementary Cementitious Material

SonReb Combined RH and UPV Method
UPV Ultrasonic Pulse Velocity
W/C Water–Cement Ratio

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Data availability statement

The data presented in this study are available upon request from the corresponding author.

CRediT authorship contribution statement

Rauf Hassan: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Udeme Udo Imoh:** Writing – review & editing, Visualization, Validation, Software. **Majid Movahedi Rad:** Writing – review & editing. **Erik Schlangen:** Writing – review & editing. **Efe Ewaen Ikponmwosa:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Mubaraq Ali:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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